FEUP

FACULTY OF ENGINEERING - UNIVERSITY OF PORTO

# Systems Engineering

Analysis and Requirements Specification

## **SWAID** - Plate Resonance (PR)

PLATE RESONANCE
TEAM


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# 1 Introduction

## 1.1 Document Presentation

This document constitutes Deliverable #2 for the Systems Engineering course. Its purpose is to detail the requirements specification and functional analysis for the **Plate Resonance (PR)** subsystem, integrated within the global **SWAID** project.

## 1.2 References

- Project Proposal 13: SWAID - Standing Waves Acoustic Interactive Demonstrator - Plate Resonance (PR).
- Systems Engineering Course Slides (FEUP).

## 1.3 Terms and Definitions

**SWAID** Standing Waves Acoustic Interactive Demonstrator.

**PR** Plate Resonance (Plate Resonance Subsystem).

**CP** Chladni Plate (Chladni Plate Subsytem)

**HI** Human Interface (Human Interface Subsystem).

**RPI** Raspberry Pi (Embedded System)

**FPGA** Field Programmable Gate Array (Hardware)

**SPI** Serial Protocol Interface

**I2C** Inter-Integrated Circuit

**UART** Universal Asynchronous Receiver-Transmitter

**API** Application Programming Interface

## 1.4 Changes

| Version | Date | Author | Description |
|---|---|---|---|
| 1 | $27^{th}$ Feb | Lucas Aparício | Template creation |
| 2 | $5^{th}$ Mar | A. Cruz, J. Oliveira, L. Aparício and M. Amorim | First Version |
| 3 | $6^{th}$ Mar | M. Amorim and L. Aparício | Second Version |

# 2 System General Perspective

## 2.1 Scope and Objectives

In accordance with Project Proposal 13, the SWAID project is an interactive standing wave demonstrator utilizing acoustic phenomena. The specific objectives pertaining to the Plate Resonance (PR) subsystem include:

- **Deterministic Resonant Mode Mapping:** Develop an extensive "dictionary" of complex resonant modes universe. This involves identifying stable standing wave frequencies and phases that produce distinct geometric symbols, and define a software abstraction layer for the Human Interface (HI) team, that translates high level user commands into digital control signals for hardware actuation.
- **High-Fidelity Signal Orchestration:** Engineer a synchronized command architecture that drives multiple acoustic transducers and lighting hardware. This ensures that Chladni symbols and synchronized illumination function as a singular, cohesive visual manifestation of the acoustic input and user interaction.
- **Latency-Critical Interaction:** Minimize the processing-to-actuation latency to ensure real-time responsiveness. The goal is to provide instantaneous feedback to both musical background modulation and hand-tracking information, maintaining the illusion of a direct physical connection between the user and the standing waves.

## 2.2 Global System Vision

As a unified final product, the SWAID demonstrator envisions an immersive, interactive scientific exhibit that harmonizes technology with art. By seamlessly integrating human gesture recognition (HI), deterministic signal orchestration (PR), and high-fidelity physical actuation (CP), the system allows users to intuitively observe the physical manifestation of sound waves in real-time.

Within this three-tiered architecture, the PR subsystem functions as the critical "translator" and synchronization engine. It bridges the high-level user interaction data from the HI subsystem with the high-performance hardware developed by the CP subsystem.

The PR team is fundamentally responsible for managing the "Resonance Modes Universe," actively synthesizing the specific frequencies and phases required to manifest stable Chladni patterns. Beyond acoustic excitation, the PR subsystem is tasked with dispatching the precise timing signals that dictate illumination behavior, ensuring that the visual and acoustic elements of the demonstrator function as a single, unified effect.

## 2.3 Functional Features

### 2.3.1 Command Interpretation & State Orchestration

The subsystem shall implement a deterministic state machine (or high-level logical architecture) to interpret incoming commands from the Human Interface (HI) team. This module defines the system's precise operating point by resolving input parameters into specific acoustic and visual states.

### 2.3.2 Deterministic Mapping of "Resonance Alphabets"

The system shall maintain a verified library—or "alphabet"—of stable resonance modes discovered verified empirically. Each logical command is mapped to a preset configuration of acoustic

frequencies and corresponding illumination states, ensuring the physical symbols and lighting remain perfectly synchronized as a singular demonstrator unit.

#### 2.3.3 Agnostic Multichannel Signal Orchestration

The subsystem provides a hardware-agnostic signal synthesis engine capable of driving multiple acoustic transducers simultaneously. It supports independent control over frequency, phase, and amplitude, and remains decoupled from the specific "Resonance Alphabet" to allow physical reconfiguration of transducers without logic redesign.

#### 2.3.4 Low-Latency State Transitions

To maintain an "interactive" and "appealing" experience, the system shall prioritize low-latency execution of state changes. Transitions between resonant modes and illumination patterns must be sufficiently fast to ensure the physical media (sand/powder) and lighting respond fluidly to the dynamics of user interaction.

#### 2.3.5 Communication Protocol Decoding & Interface Abstraction

In the FPGA-based architecture, the subsystem shall implement a dedicated hardware-level decoder to process signals from the compact computer. This feature abstracts the complexity of the electrical communication protocol (e.g., SPI, I2C, or UART), extracting relevant metadata to feed the primary state machine for real-time actuation.

### 2.4 User Characteristics

The SWAID demonstrator is designed for three primary categories of users, each with distinct interaction requirements:

- **General Public:** Non-technical users at scientific outreach events. They require an intuitive, "hands-off" interface where interaction is limited to hand gestures or music selection. The system must provide high-attractiveness visual feedback with minimal learning curve.
- **Scientific Presenters:** Educators who utilize the system for formal demonstrations. These users require a "plug-and-play" setup that remains stable during continuous operation and allows for a repeatable sequence of Chladni patterns.
- **System Maintenance (Internal Teams):** Technical users (HI, PR, and CP teams) who interact with the software abstraction layer. They require the ability to reconfigure the "Resonance Alphabet" and perform system calibrations via the RPI command line or FPGA interface.

### 2.5 Inter-Team Dependencies:

- **Acoustic Characterization (PR $\leftarrow$ CP):** The PR subsystem's ability to map the "Resonance Alphabet" is strictly dependent on the physical properties (material, thickness, and mounting) of the plate provided by the CP team. Any structural changes to the plate will require a full recalibration of the PR dictionary.
- **Settling Time vs. Command Rate (PR $\leftarrow$ HI):** While the PR electronic system can switch signals in milliseconds, the physical inertia of the powder requires a settling time to form stable images. *Mitigation:* The HI team must implement a "state-hold" or "debounce" logic

in their software to prevent switching symbols faster than the physical medium can respond.

- **Gesture-to-Symbol Latency (PR ← HI):** The "Real-Time" attractiveness of the demonstrator relies on the HI team's image processing speed. The PR team's low-latency performance (PR-05) is only effective if the HI team provides modulation commands with minimal pre-processing delay.
- **Interface Contract (PR ⇔ HI):** The PR and HI teams must mutually define the software development stack and the data exchange format early in the development cycle. The PR team is dependent on the HI team adhering to the established API definitions to ensure successful integration of the "Resonance Alphabet" commands. Unilateral changes to this interface by either team will compromise the subsystem's integration.

# 3 Requirements and Constraints Table

Table 1: SWAID Plate Resonance (PR) Requirements Specification

| ID | Type | Priority | Description |
|---|---|---|---|
| PR-01 | Functional | High | The PR subsystem shall generate independent actuation signals (frequency, phase, and amplitude) for multiple acoustic transducers to produce stable Chladni patterns. |
| PR-02 | Functional | High | The subsystem shall implement an internal reference library that directly translates high-level system inputs (e.g., target geometric patterns or user gesture commands) into the corresponding transducer parameters (frequency, phase and amplitude). |
| PR-03 | Functional | Medium | The PR subsystem shall generate illumination control signals that are synchronized with the active Chladni resonance mode state. |
| PR-04 | Performance | High | The processing-to-actuation latency, from receiving an HI command to updating transducer and lighting outputs, shall be sufficiently low to present a unified, real-time response without visually apparent discrepancies. |
| PR-05 | Performance | Medium | The subsystem shall ensure tight temporal alignment between the acoustic output and the lighting control commands, avoiding any visually discernible delay between pattern formation and illumination changes. |
| PR-06 | Design Constraint | Medium | The signal generation architecture (FPGA/Drivers) shall be Alphabet-Agnostic, supporting dynamic remapping of resonance modes to physical transducers via configuration parameters without requiring a redesign of the core logic. |

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**Table 1 – Continued from previous page**

| ID | Type | Priority | Description |
|---|---|---|---|
| PR-07 | Design Constraint | High | The PR subsystem shall digitally limit the maximum amplitude and duty cycle of the output signals to prevent exceeding the thermal/power ratings of the CP transducers. |
| PR-08 | Design Constraint | Medium | The physical dimensions of the PR subsystem's computational hardware shall fit entirely within the structural footprint enclosure designated by the CP team. |
| PR-09 | External Interface | High | The PR subsystem shall provide a software abstraction layer (API) to the HI team to accept high-level command strings representing musical notes, chords, or gestures. |
| PR-10 | Internal Interface | High | The PR subsystem shall transmit translated actuation commands to the signal generator (FPGA or sound card) using a standard digital communication protocol (e.g., SPI, UART). |
| PR-11 | Quality Attribute | Medium | The PR subsystem software/firmware shall support continuous operation for a minimum of 8 hours without requiring a system reboot or experiencing state failure. |
| PR-12 | Quality Attribute | Low | The PR subsystem shall automatically load the resonance dictionary and enter a "ready-to-receive" state within 10 seconds of system power-on. |